DisasterWatch — Real-Time Disaster Monitoring System

System Architecture, Real-Time Telemetry Lineage, and Risk Engine Methodology

MANDATORY SAFETY DISCLAIMER:
"This student project is for disaster awareness and monitoring only. Always follow official instructions from local authorities, disaster-management agencies, and emergency services. It never claims to control emergency services, dispatch help, or replace official disaster-management authorities."

1. Project Purpose & Scope

DisasterWatch is a global, web-based disaster situational awareness and decision-support system. The project aggregates authentic, instrumental telemetry from international scientific agencies—including the United States Geological Survey (USGS), Open-Meteo High-Resolution Numerical Weather Prediction Models, and the United Nations / European Commission Global Disaster Alert and Coordination System (GDACS). The platform eliminates synthetic data, providing genuine coordinates, verifiable timestamps, and transparent risk calculations.

2. Event Classification Taxonomy

To ensure scientific accuracy and prevent public misinterpretation, events are classified into three distinct categories:

Classification	Visual Representation	Scientific Definition & Source
Confirmed Event	Solid Marker	Direct instrumental sensor detections (e.g. USGS seismometer arrivals, VIIRS satellite inf
Official Alert	Solid Boundary / Circle	Statutory regulatory warnings issued by sovereign or international emergency bodies (e.g
Estimated Risk Zone	Dashed Outline (Low Opacity)	Calculated perimeter derived by the Python risk engine from live atmospheric telemetrics

3. Mathematical Methodology: Live Telemetric Disaster Level Calculation

Disaster levels (Low, Medium, High, Critical) are computed through deterministic physical transfer functions operating on streaming live telemetrics. Each hazard type evaluates measurable physical variables against standard geophysical benchmarks:

3.1 Seismic Hazard Intensity (USGS Live Stream)

Seismic severity is evaluated using reported Moment Magnitude (M) combined with a focal hypocentral depth ground attenuation factor (f_depth):

f_depth = 1.0 if Depth ≤ 30 km (Shallow Crustal) else 0.82 (Intermediate / Deep)

Seismic Risk Index (SRI) = min(10.0, [2.0 + (M / 4.0) * 1.5] * f_depth) for M < 4.0

SRI = min(10.0, [4.0 + ((M - 4.0) / 2.0) * 2.5] * f_depth) for 4.0 ≤ M < 6.0

SRI = min(10.0, [6.8 + ((M - 6.0) / 1.0) * 1.7] * f_depth) for 6.0 ≤ M < 7.0

SRI = min(10.0, [8.5 + (M - 7.0) * 0.75]) for M ≥ 7.0 (Critical)

3.2 Estimated Flood Risk (Open-Meteo Atmospheric Models)

Flood potential is determined by combining short-duration burst rainfall intensity (I_h, mm/h) and 24-hour antecedent cumulative precipitation (P_24, mm):

Hydrological Inundation Potential (HIP) = (0.65 * (I_h / 30.0) + 0.35 * (P_24 / 100.0)) * 10.0

- Low Risk: I_h ≤ 5 mm/h AND P_24 ≤ 20 mm (HIP < 3.0)
- Medium Risk: I_h > 5 mm/h OR P_24 > 20 mm (3.0 ≤ HIP < 5.5)
- High Risk: I_h > 15 mm/h OR P_24 > 50 mm (5.5 ≤ HIP < 8.0)
- Critical Risk: I_h > 30 mm/h OR P_24 > 100 mm (HIP ≥ 8.0)

3.3 Tropical Cyclone & Gale Wind Shear (GDACS & Open-Meteo)

Evaluates 10-meter sustained wind velocity (V_10, km/h) and peak wind gusts (G_max) using IMD and Saffir-Simpson thresholds:

Effective Wind Kinetic Hazard (WKH) = 0.5 * ρ_air * (V_10 / 3.6)² (Dynamic Wind Pressure)

- Low: V_10 < 31 km/h (Gentle/Moderate breeze)
- Medium (Depression): 31 ≤ V_10 < 62 km/h
- High (Cyclonic Storm / Gale): 62 ≤ V_10 < 89 km/h (Trees uprooted, structural warnings)
- Critical (Severe Cyclonic Storm / Hurricane): V_10 ≥ 89 km/h (Widespread destruction)

3.4 Heatwave & Human Heat Index (NOAA / IMD Formulation)

Calculates apparent thermal stress from 2-meter ambient temperature (T, °C) and relative humidity (RH, %):

Heat Index (HI) = -8.784 + 1.611*T + 2.338*RH - 0.146*T*RH - 0.0123*T² - 0.0164*RH² + 0.0022*T²*RH

- Low: T < 38 °C (Normal conditions)
- Medium: 38 ≤ T < 42 °C (Pre-heatwave caution)
- High: 42 ≤ T < 45 °C (Official Heatwave advisory)
- Critical: T ≥ 45 °C OR HI ≥ 54 °C (Severe Heatwave / Extreme medical risk)

3.5 Active Wildfire Thermal Intensity (NASA FIRMS / GDACS)

Satellite thermal radiative energy is measured through Fire Radiative Power (FRP, Megawatts) and 4-micron brightness temperature (T_b4, Kelvin):

- Low: FRP ≤ 10 MW (Small agricultural burn / nascent heat spot)
- Medium: 10 < FRP ≤ 30 MW (Active surface combustion)
- High: 30 < FRP ≤ 100 MW (Intense wildfire front)
- Critical: FRP > 100 MW OR T_b4 > 360 K (Crown fire / Mega-fire front)

4. Standardized Emergency Safety Protocols

Hazard	Immediate Action Directive
Earthquake	Drop, Cover, and Hold On; move away from damaged structures after shaking ceases.
Flood	Avoid walking or driving through flood water; evacuate to higher ground upon authority advisory.
Fire	Follow local evacuation corridors, shut gas valves, and avoid particulate smoke inhalation.
Cyclone	Remain indoors away from glass windows; heed maritime storm surge evacuation orders.
Heatwave	Maintain hydration, avoid sun exposure between 11:00-16:00, and monitor vulnerable individuals.

5. Recommended Future Enhancements

1. **Global Gridded Population Exposure:** Overlay LandScan or GPWv4 population density raster tiles to dynamically estimate the number of inhabitants within a 50km hazard radius.
2. **Historical Replay Slider:** Implement a temporal 7-day replay control to observe storm progression and seismic aftershock sequences.
3. **Multi-Sensor Tsunami Buoy Telemetry:** Ingest NOAA DART (Deep-ocean Assessment and Reporting of Tsunamis) acoustic sensor readings.
4. **Offline Progressive Web App (PWA):** Cache critical offline safety checklists and local map tiles via Service Workers for low-connectivity disaster zones.
5. **Automated Webhook / MQTT Alert Dispatch:** Allow municipal emergency centers to subscribe to machine-readable alert webhooks.